

Chapter 6: Distribution of Sampling Statistics

ST2334 Probability and Statistics¹
(Academic Year 2014/15, Semester 1)

Department of Statistics & Applied Probability (DSAP)
National University of Singapore (NUS)¹

Outline

- 1 Introduction
- 2 The Sample Mean
- 3 The Central Limit Theorem
- 4 The Sample Variance
- 5 Sampling Distributions From a Normal Population

Introduction I

Learning Outcomes

Questions to Address:

- ◆ View population & sample in a statistical problem as r.v.'s
- ◆ Mean & variance of the sample mean
- ◆ Using the empirical method to find a sampling distribution
- ◆ Using the central limit theorem to approximate the distribution of a sum/an average of i.i.d. r.v.'s
- ◆ Normal approximation to the binomial distribution with continuity correction
- ◆ Mean of the sample variance
- ◆ Some sampling distributions from a normal population

Introduction II

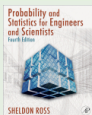
Learning Outcomes (continued)

Concept & Terminology:

- ◆ underlying distribution
- ◆ parametric/nonparametric inference problem
- ◆ statistic/sampling statistic
- ◆ sampling distribution of a statistic
- ◆ sample sum/mean/variance/standard deviation
- ◆ population mean/variance
- ◆ empirical method
- ◆ sampling distribution of the sample sum/mean/variance
- ◆ central limit theorem
- ◆ normal approximation to the binomial distribution with continuity correction
- ◆ joint sampling distribution of the sample mean & the sample variance from a normal population
- ◆ t -statistic constructed from a normal population

Introduction III

Mandatory Reading



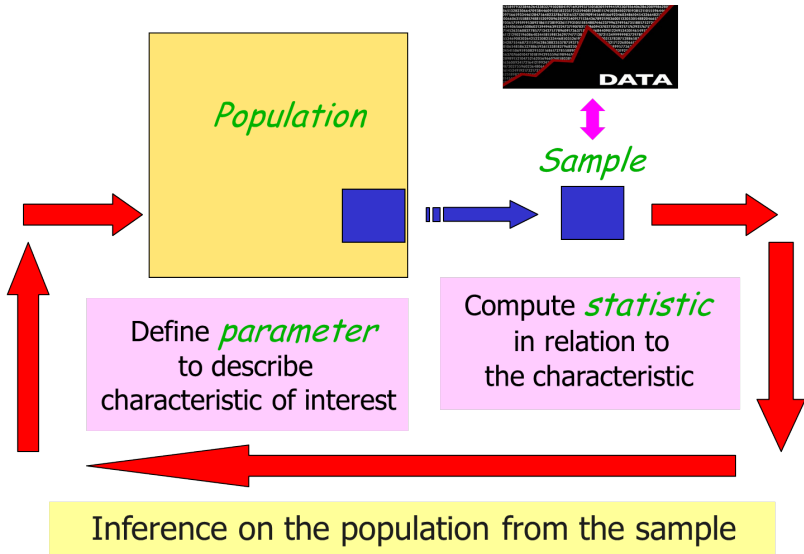
➡ Section 6.1 – Section 6.5

**The science of statistics deals with
drawing conclusions from observed data**

▶ **Recall** (in Ch. 0) the set-up of a *statistical problem*:

- ▶ *population*: a large collection of items (each is associated with some values/measurements)
- ▶ *sample/data*: only a subset of the population (*i.e.*, part of all measurements) is available

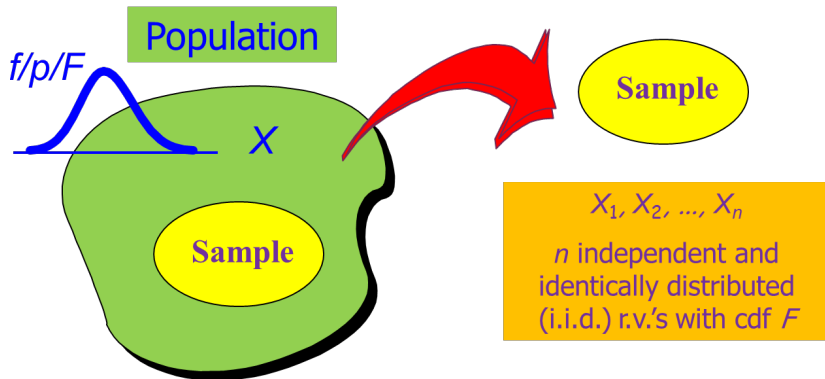
Introduction IV



Set-up of Statistical Problems Via r.v.'s

Idea :

- ▶ *population* $\Leftrightarrow$ r.v. X with *unknown* cdf F (& density f/p)
- ▶ *sample/data* $\Leftrightarrow$ i.i.d. r.v.'s, $X_1, \dots, X_n$, with cdf F (& density f/p)
- ▶ Refer to the distribution of X or the cdf F as the underlying distribution



Parametric Inference Problems I

- ▶ In practice, we often select a particular parametric family of probability distributions parametrized by some parameters θ as the underlying distribution F of the population of interest, e.g.,
- 1 Lifetime of a light bulb : an *exponential* r.v.
 - 2 annual revenue of Singapore Pools : a *normal* r.v.
 - 3 # of calls  in an hour at a pizza delivery shop: a *Poisson* r.v.
- ▶ Choosing with care appropriate values for θ allows us to model/deal with most random phenomena or population of interest
- ▶ It suffices to select values for θ in the particularly selected family, i.e.,
- 1 $\theta = \lambda \in (0, \infty)$ in the exponential case
 - 2 $\theta = (\mu, \sigma^2) \in \mathbb{R} \times (0, \infty)$ in the normal case
 - 3 $\theta = \lambda \in (0, \infty)$ in the Poisson case

Parametric Inference Problems II

Definition

A parametric inference problem has a set-up in which the underlying distribution of the population of interest is assumed to be from a particular *parametric family of probability distributions* which is *parametrized by some parameters θ*

- ▶ When there is NOT any assumption made on the form of the cdf F , such a problem is called a nonparametric inference problem
- ▶ **Note**: This module *ONLY* looks at parametric inference problems

Sampling Statistics

- ▶ **Recall** (in [Example 4 \(Ch.0\)](#)): In understanding p , the unknown proportion of the Singaporean population  who have a  a/c (parameter of interest $\theta = p$), we compute & study the proportion of people who have a  a/c among a group of  students (a sample)

Definition

A *sampling statistic (or simply statistic)* summarizes relevant information from a sample in helping us to understand the population or the underlying distribution

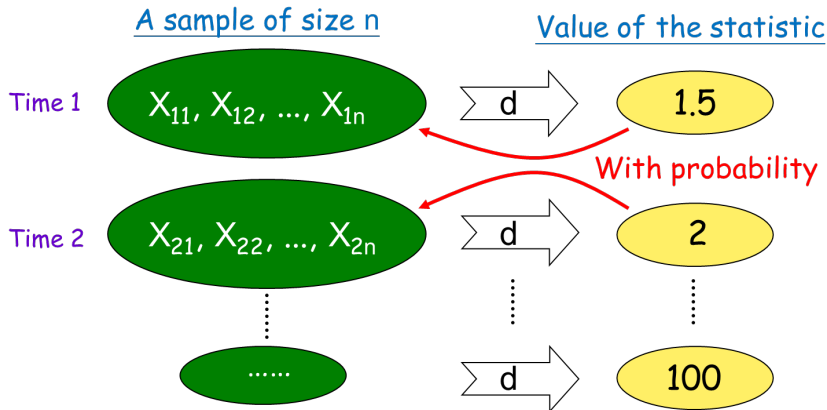
Definition

In *parametric inference problems*, a *sampling statistic (or simply statistic)* summarizes relevant information to infer/guess the true but unknown value of the parameter θ of the underlying distribution

Sampling Statistics is a r.v.

Statistic as a r.v.

A statistic, defined by $d(X_1, \dots, X_n)$, is a *r.v.* with possible values governed by the data/sample (i.e., $X_1, \dots, X_n$) through some *known/pre-determined function* d of n i.i.d. r.v.'s



Sampling Distribution I

Definition

The *probability distribution of a statistic* is called the sampling distribution

In practice, we observe

- ▶ n known values, $x_1, \dots, x_n$, as n data points from a sample wherein $X_1 = x_1, X_2 = x_2, \dots, X_n = x_n$ are n i.i.d. realizations from the underlying distribution
- ▶ *ONLY 1 single value* $d(x_1, \dots, x_n)$ called the observed value as a realization/draw from the sampling distribution of a statistic

Sampling Distribution II

Suppose that we obtain 1.5 as the value of a statistic:

How to interpret this number?

*Is this number a good guess of the
unknown value of the parameter?*

Close to it?

Address these questions by
understanding the sampling distribution !

The Sample Mean $\bar{X}$ I

Recall (in [Example 19 \(Ch.4\)](#)): We define the sample mean $\bar{X}$ there. Here is an *alternative definition*

Definition

- 1 A population of interest whose members or values can be regarded as being the values of a r.v. X with cdf F
- 2 The mean $E(X) = \mu$ & variance $\text{Var}(X) = \sigma^2$ are called the *population mean* & the *population variance*, respectively
- 3 Let $X_1, \dots, X_n$ be a sample from the population, i.e., *n i.i.d. r.v.'s* having cdf F
- 4 The sample mean is defined by

$$\bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n} = \frac{1}{n} \sum_{i=1}^n X_i \quad (1)$$

The Sample Mean $\bar{X}$ II

▶ $\bar{X}$ is **1 of the most commonly used statistics** as it

- 1 is defined as a function d of n i.i.d. r.v.'s from (1): d is the average function
- 2 serves as the most appropriate statistic which helps to *estimate/guess the unknown value of the mean μ of ANY population*

Mean & Variance of The Sample Mean $\bar{X}$

The sample mean $\bar{X}$ is a r.v. with *mean* & *variance*

$$E(\bar{X}) = \mu \quad \& \quad \text{Var}(\bar{X}) = \frac{\sigma^2}{n}$$

- ▶ Shown in Examples 19 & 26 (Ch.4)
- ▶ $\bar{X}$ always has a “*centre*” at the population mean μ for all sample size n
- ▶ The *spread* of $\bar{X}$ ↓ as the sample size n ↑

Sampling Distribution of $\bar{X}$

- ▶ The true sampling distribution of $\bar{X}$ is obtainable through *vigorous prob computations* that are rather tedious

This module would not consider it

Empirical Method

In principle, the sampling distribution of $\bar{X}$ (or, of any statistic) can be obtained by the empirical method (depicted at the next page):

- ➊ Collect *ALL possible samples of size n* from the population X
- ➋ Compute a value $\bar{x}$ from each sample in ➊
- ➌ Construct a *relative frequency histogram* of all $\bar{x}$ values in ➋

- ▶ The # of all possible samples of size n in ➊ can be *astonomical* or even *uncountable infinite*
- ▶ An approximate sampling distribution of $\bar{X}$: Collect *a large # of samples of size n* in ➊, followed by ➋ & ➌

The Empirical Method

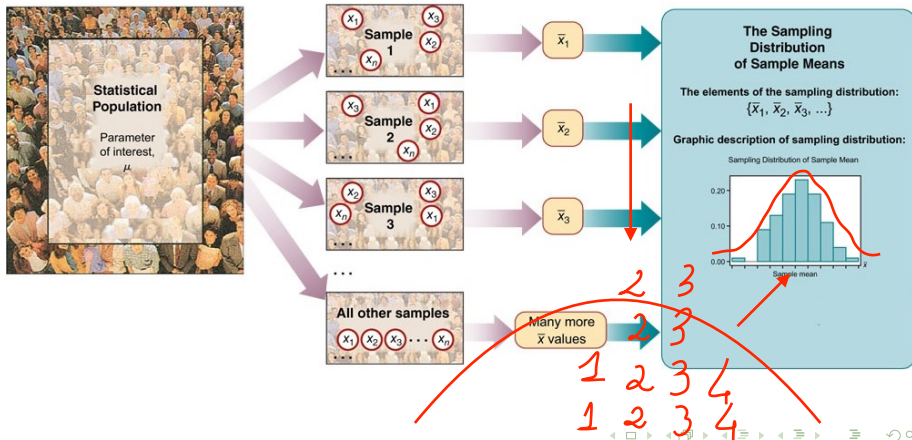
**Statistical
population
being
studied**

Repeated sampling is needed to form the sampling distribution

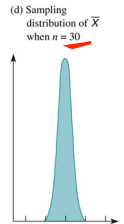
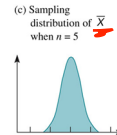
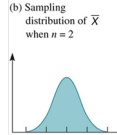
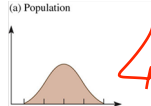
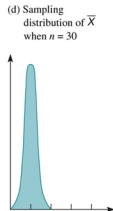
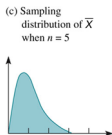
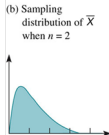
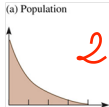
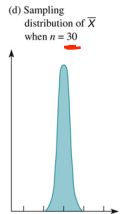
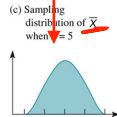
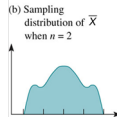
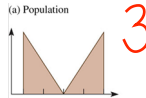
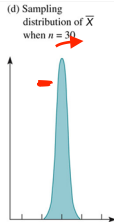
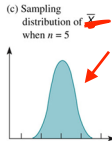
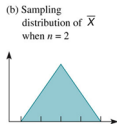
All possible
samples
of size n

One value of the sample statistic ($\bar{x}$ in this case) corresponding to the parameter of interest (μ in this case) is obtained from each sample

Then all of these values of the sample statistic, $\bar{x}$, are used to form the sampling distribution

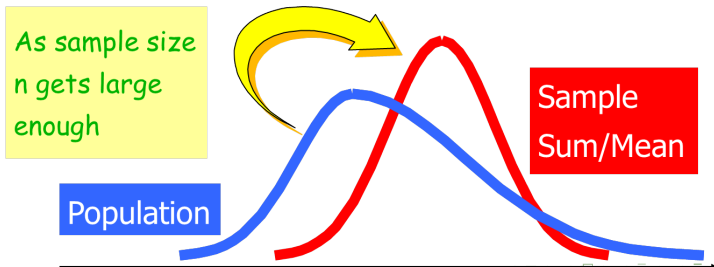


Example 1: Sampling Distributions of $\bar{X}$



Central Limit Theorem I

- ▶ The central limit theorem (CLT) is one of the *most remarkable results in probability theory*
- ▶ Loosely put, it states that the sum/average of a large # of indept r.v.'s has a distribution that is approximately normal
- ▶ It *not only* provides a simple method for computing approximate probs for sums of indept r.v.'s, but it *also* helps explain the remarkable fact that the empirical frequencies of so many natural populations exhibit bell-shaped (*i.e.*, normal) curves



Central Limit Theorem II

Central Limit Theorem (CLT)

Let $X_1, X_2, \dots, X_n$ be a sequence of i.i.d. r.v.'s having mean μ & variance σ^2 . Then for $n \geq 30$, the distribution of the sample sum

$$X_1 + X_2 + \dots + X_n = \sum_{i=1}^n X_i$$

is *approximately normal with mean $n\mu$ & variance $n\sigma^2$* . Write

$$\sum_{i=1}^n X_i \sim N(n\mu, n\sigma^2)$$

► $\frac{\sum_{i=1}^n X_i - n\mu}{\sqrt{n\sigma^2}} \sim N(0, 1) \Rightarrow P\left(\frac{\sum_{i=1}^n X_i - n\mu}{\sqrt{n\sigma^2}} < x\right) \approx \Phi(x)$

i.e., prob of a sum of i.i.d. r.v.'s can be *approximated by a standard normal prob when n is large*

Approximate Sampling Distribution of $\bar{X}$

Approximate Sampling Distribution of $\bar{X}$

Let $X_1, X_2, \dots, X_n$ be a sequence of i.i.d. r.v.'s having mean μ & variance σ^2 . Then for $n \geq 30$, the distribution of

$$\bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n} = \frac{1}{n} \sum_{i=1}^n X_i$$

is *approximately normal with mean μ & variance σ^2/n* . Write

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

- ▶ Translation of *approximate normality* through linear transformation:
For $X \sim N(\mu, \sigma^2)$, $Y = a + bX \sim N(a + b\mu, b^2\sigma^2)$ for fixed constants a & b

Example 2: Central Limit Theorem

If a fair die is rolled 30 times, find the approximate prob that the sum obtained is between 100 & 110

Solution: Let X_i be the # obtained at the i th roll of the fair die, for $i = 1, 2, \dots, 30$. We have $\mu = E(X_i) = 7/2$, $E(X_i^2) = 91/6$, & therefore $\sigma^2 = \text{Var}(X_i) = 35/12$. The required prob is

$$P(100 \leq X_1 + \dots + X_{30} \leq 110)$$

Apply CLT based on $X_1 + \dots + X_{30} \sim N(n\mu, n\sigma^2) = N\left(30 \times \frac{7}{2}, 30 \times \frac{35}{12}\right)$:

$$\begin{aligned} & P(100 \leq X_1 + \dots + X_{30} \leq 110) \\ & \approx P\left(\frac{100 - 30 \times (7/2)}{\sqrt{30 \times (35/12)}} \leq Z \leq \frac{110 - 30 \times (7/2)}{\sqrt{30 \times (35/12)}}\right) \\ & = P(-.53 \leq Z \leq .53) \\ & = .4038 \end{aligned}$$

Example 3: Central Limit Theorem I

The # of students who enrol in a psychology class is a Poisson r.v. with mean 100. The professor in charge of the course decided that if the number of enrollment is ≥ 120 , he will teach the course in 2 separate sessions, whereas if the enrollment is under 120 he will teach all the students in a single session. What is the prob that the professor will have to teach 2 sessions?

Solution: Let X be the enrollment in the psychology class. Given that $X \sim \text{Poi}(100)$ with $E(X) = 100 = \text{Var}(X)$. The required prob,

$$P(X \geq 120) = e^{-100} \sum_{i=120}^{\infty} \frac{100^i}{i!},$$

is not readily available by hand

Remark: Using a simple spreadsheet in computer, this prob is computed to be $1 - .9718 = .0282$

Example 3: Central Limit Theorem II

Alternatively, realize that

$$X = X_1 + \cdots + X_{100}$$

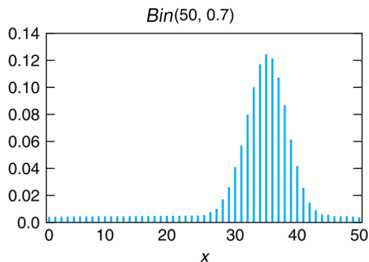
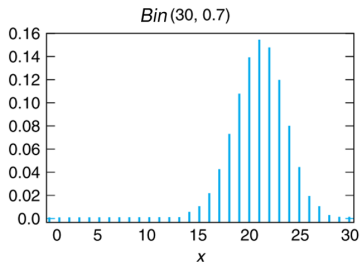
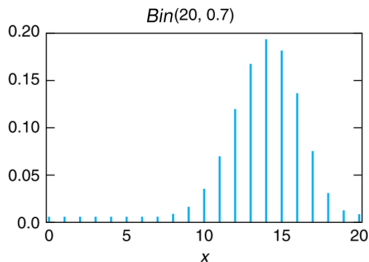
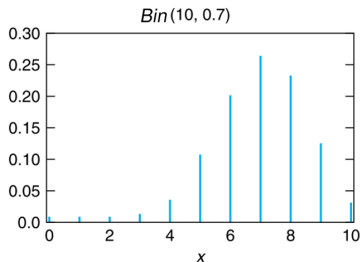
where X_i are i.i.d. $Poi(1)$ r.v.'s with $\mu = \sigma^2 = 1$, & apply CLT to conclude that

$$X \sim N(n\mu = 100, n\sigma^2 = 100) \quad \Rightarrow \quad \frac{X - 100}{\sqrt{100}} \sim N(0, 1)$$

Then,

$$\begin{aligned} P(X \geq 120) &= P\left(\frac{X - 100}{\sqrt{100}} \geq \frac{120 - 100}{\sqrt{100}}\right) \\ &\approx P\left(Z \geq \frac{20}{\sqrt{100}}\right) \\ &= P(Z \geq 2) \\ &= .0228 \end{aligned}$$

The Normal Approximation to the Binomial Distribution I



Binomial pmf converging to a bell-shaped curve when $n \uparrow$

The Normal Approximation to the Binomial Distribution II

▶ Normal approximation to the binomial distribution

- ▶ 1 extremely important application of CLT: A $\text{Bin}(n, p)$ r.v. is a sum of n i.i.d. $\text{Ber}(p)$ r.v.'s when the *underlying distribution is a Bernoulli r.v.*
- ▶ such an approximation by CLT is generally quite good for values of n *satisfying $npq \geq 10$* , & it will be further improved if we incorporate continuity correction (cc)

Normal Approximation to Binomial Probabilities With Continuity Correction

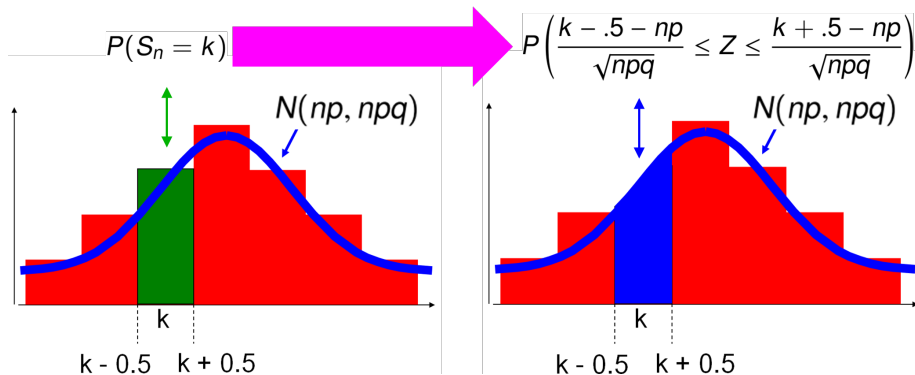
If $S_n \sim \text{Bin}(n, p)$ with $q = 1 - p$, then, for $k = 0, 1, \dots, n$,

$$\begin{aligned} P(S_n = k) &= P(k - .5 \leq S_n \leq k + .5) \\ &\approx P\left(\frac{k - .5 - np}{\sqrt{npq}} \leq Z \leq \frac{k + .5 - np}{\sqrt{npq}}\right) \end{aligned}$$

▶ $P(S_n \geq k) = P(S_n \geq \underline{k - .5})$ & $P(S_n \leq k) = P(S_n \leq k + .5)$

The Normal Approximation to the Binomial Distribution III

Normal Approximation With Continuity Correction



Example 4: Normal Approximation

Let X be a binomial r.v. with parameters 60 & .3

$$\begin{aligned} \textcircled{1} \quad P(12 \leq X \leq 26) &\approx P\left(\frac{11.5 - 18}{\sqrt{12.6}} \leq Z \leq \frac{26.5 - 18}{\sqrt{12.6}}\right) \\ &\approx P(-1.83 \leq Z \leq 2.39) = .9916 - (1 - .9664) = .9580 \end{aligned}$$

$$\begin{aligned} \textcircled{2} \quad P(12 < X \leq 26) &= P(13 \leq X \leq 26) \\ &\approx P\left(\frac{12.5 - 18}{\sqrt{12.6}} \leq Z \leq \frac{26.5 - 18}{\sqrt{12.6}}\right) \approx P(-1.55 \leq Z \leq 2.39) \\ &= .9916 - (1 - .9394) = .9310 \end{aligned}$$

$$\begin{aligned} \textcircled{3} \quad P(12 \leq X < 26) &= P(12 \leq X \leq 25) \\ &\approx P\left(\frac{11.5 - 18}{\sqrt{12.6}} \leq Z \leq \frac{25.5 - 18}{\sqrt{12.6}}\right) \approx P(-1.83 \leq Z \leq 2.11) \\ &= .9826 - (1 - .9664) = .9490 \end{aligned}$$

$$\begin{aligned} \textcircled{4} \quad P(12 < X < 26) &= P(13 \leq X \leq 25) \\ &\approx P\left(\frac{12.5 - 18}{\sqrt{12.6}} \leq Z \leq \frac{25.5 - 18}{\sqrt{12.6}}\right) \approx P(-1.55 \leq Z \leq 2.11) \\ &= .9826 - (1 - .9394) = .9220 \end{aligned}$$

Example 5: Normal Approximation

Let X be the # of times that a fair coin, flipped 40 times, lands heads. Suppose that we are interested in seeing heads half of the time. Compute the prob exactly & approximate it

Solution: *Exact answer:* $P(X = 20) = \binom{40}{20} \left(\frac{1}{2}\right)^{40} = .1254$

Normal approximation:

Handwritten notes: The term $\left(\frac{1}{2}\right)^{40}$ is annotated with a red bracket underneath, and the number 40 is written in red. To the right of the equation, the fraction $\frac{1}{2}$ is written twice in red, with a red arrow pointing from the bracketed term to the first $\frac{1}{2}$.

$$\begin{aligned} P(X = 20) &= P(19.5 \leq X \leq 20.5) \\ &= P\left(\frac{19.5 - 20}{\sqrt{10}} \leq \frac{X - 20}{\sqrt{10}} \leq \frac{20.5 - 20}{\sqrt{10}}\right) \\ &\approx P(-0.16 \leq Z \leq 0.16) = .1272 \end{aligned}$$

Example 6: Normal Approximation

The ideal size of a first-year class at a particular college is 150 students. The college, knowing from the past experience that on the average only 30% of those accepted for admission to this class will actually attend, uses a policy of approving the applications of 450 students. Compute the prob that > 150 students attend this class

Solution: Let X denote the # of students that attend among the 450 students. Then, $X \sim \text{Bin}(450, .3)$ with mean & variance

$$E(X) = 450 \times .3 = 135 \quad \& \quad \text{Var}(X) = 450(.3)(.7) = 9.721^2$$

Applying normal approximation with cc, the prob that > 150 students attend this class is

$$\begin{aligned} P(X > 150) &= P(X \geq 150.5) \approx P\left(Z \geq \frac{150.5 - 135}{9.721}\right) \\ &= P(Z \geq 1.59) = .0559 \end{aligned}$$

Example 7: Normal Approximation I

Refer to the previous example, if this college desires that the prob of > 150 students will attend this college should be at most .01. What is the largest # of students should this college admit?

Solution: Let X denote the # of students that attend. Let n be the # of students to be admitted. Then, $X \sim \text{Bin}(n, .3)$ with mean & variance

$$E(X) = .3n \quad \& \quad \text{Var}(X) = .3n \times .7 = .21n$$

Applying normal approximation with cc yields the prob that > 150 students will attend the college,

$$\begin{aligned} P(X > 150) &= P(X \geq 150.5) \\ &\approx P\left(Z \geq \frac{150.5 - .3n}{\sqrt{.21n}}\right) \end{aligned}$$

Example 7: Normal Approximation II

To have this probability to be at most .01, set it to be less than or equal to .01. This yields

$$P\left(Z \geq \frac{150.5 - .3n}{\sqrt{.21n}}\right) \leq P(Z \geq 2.33) = \underline{.01}$$

Accordingly

$$\frac{150.5 - .3n}{\sqrt{.21n}} \geq 2.33 \quad \Rightarrow \quad n \leq 428.32$$

Hence, the largest # of students this college should admit is 428

The Sample Variance I

Definition

Let $X_1, \dots, X_n$ be i.i.d. r.v.'s having mean μ & variance σ^2 , & $\bar{X} = \sum_{i=1}^n X_i/n$ be the **sample mean**. The sample variance is a statistic defined by

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

$S = +\sqrt{S^2}$ is called the sample standard deviation

Mean of the Sample Variance Equals the Population Variance σ^2

The expected value/mean of the sample variance S^2 always equals σ^2 :

$$E(S^2) = \sigma^2$$

- Serves as an appropriate statistic in **estimating the unknown value of the variance of ANY population**

The Sample Variance II

- Start with the following *important algebraic identity of S^2* :

$$\begin{aligned}(n-1)S^2 &= \sum_{i=1}^n (X_i - \mu + \mu - \bar{X})^2 \\&= \sum_{i=1}^n (X_i - \mu)^2 + \sum_{i=1}^n (\bar{X} - \mu)^2 - 2(\bar{X} - \mu) \sum_{i=1}^n (X_i - \mu) \\&= \sum_{i=1}^n (X_i - \mu)^2 + n(\bar{X} - \mu)^2 - 2(\bar{X} - \mu)n(\bar{X} - \mu) \\&= \sum_{i=1}^n (X_i - \mu)^2 - n(\bar{X} - \mu)^2\end{aligned}$$

Taking expectation of both sides yields

$$\begin{aligned}(n-1)E(S^2) &= \sum_{i=1}^n E[(X_i - \mu)^2] - nE[(\bar{X} - \mu)^2] \\&= n\sigma^2 - n\text{Var}(\bar{X}) = (n-1)\sigma^2 \quad \Rightarrow \quad E(S^2) = \sigma^2\end{aligned}$$

Sampling Distributions From a Normal Population I

Let's look at a **special case** in which the underlying distribution is normal

Sampling Distribution of $\bar{X}$ From a Normal Population

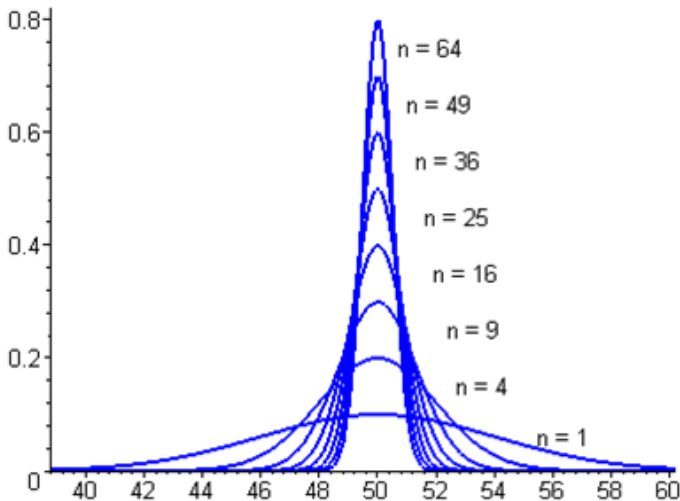
Let $X_1, \dots, X_n$ be a sample from a **normal population** with mean μ & variance σ^2 . Then,

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right) \quad \text{or} \quad \frac{\bar{X} - \mu}{\sigma / \sqrt{n}} \sim N(0, 1) \quad (2)$$

- ▶ **Valid** for all sample sizes $n = 1, 2, 3, \dots$
- ▶ Due to “linear transformation” result at **Page 29 (Ch.5)**: For $X \sim N(\mu, \sigma^2)$, $Y = a + bX \sim N(a + b\mu, b^2\sigma^2)$ for fixed constants a & b

Sampling Distributions From a Normal Population II

Sampling distribution of sample means from $N(50, 4^2)$:



Sampling Distributions From a Normal Population III

Joint Sampling Distribution of $\bar{X}$ & S^2 From a Normal Population

For a sample $X_1, \dots, X_n$ from a **normal population** with mean μ & variance σ^2 , the joint sampling distribution of $(\bar{X}, S^2)$ is described as follows:

- ① $\bar{X}$ & S^2 are indept r.v.'s
- ② $\bar{X}$ has a distribution given in (2)
- ③ $\frac{(n-1)}{\sigma^2} S^2 \sim \chi_{n-1}^2$

▶ A brief discussion of ①:

- ▶ S^2 is a function of all n deviations from $\bar{X}$, $X_i - \bar{X}$, which are all normally distributed r.v.'s
- ▶ $\text{Cov}(X_i - \bar{X}, \bar{X}) = 0$ from Examples 25 (Ch.2)
⇒ indep of $X_i - \bar{X}$ & $\bar{X}$ ($\because$ normality of the 2 r.v.'s)

Sampling Distributions From a Normal Population IV

► A streamline proof of ③:

► Note that

$$W = \frac{1}{\sigma^2} \sum_{i=1}^n (X_i - \mu)^2 = \sum_{i=1}^n \left(\frac{X_i - \mu}{\sigma} \right)^2 \sim \chi_n^2$$

& that W can be re-expressed as

$$\frac{1}{\sigma^2} \sum_{i=1}^n [(X_i - \bar{X}) + (\bar{X} - \mu)]^2 = \frac{(n-1)}{\sigma^2} S^2 + \left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \right)^2 = U + V$$

by expanding the square in the LHS & using the fact that

$$\sum_{i=1}^n (X_i - \bar{X}) = 0$$

► U & V are indept ($\because S^2$ & $\bar{X}$ are indept)

► $W \sim \chi_n^2$ & $V \sim \chi_1^2 \Rightarrow U \sim \chi_{n-1}^2$

Sampling Distributions From a Normal Population V

A t -Statistic Constructed by $\bar{X}$ & S^2 From a Normal Population

Let $\bar{X}$ & S^2 be the sample mean & the sample variance of a sample from a *normal population* with mean μ & variance σ^2 . Then,

$$\frac{\bar{X} - \mu}{\sqrt{S^2/n}} \sim t_{n-1} \quad (3)$$

- Follows from definition of a t -distribution by expressing the ratio as

$$\frac{\bar{X} - \mu}{\sqrt{S^2/n}} = \frac{\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \right)}{\sqrt{S^2/\sigma^2}} = \frac{\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \right)}{\sqrt{[(n-1)S^2/\sigma^2]/(n-1)}}$$

where the r.v. at the numerator is $N(0, 1)$ & the r.v. at the denominator is $\sqrt{\chi_{n-1}^2/(n-1)}$, & also the 2 r.v.'s are indept

Example 8: Sampling Distributions From a Normal Population I

Suppose that the first monthly salary (in S\$) of a undergraduate in a university is known to be approximately $N(2000, 900)$. For a selected group of 15 students, what are the probs that

- 1 the average first monthly salary is greater than S\$2010?
- 2 the standard deviation of their first monthly salaries is greater than S\$10?

Solution: Let $X_1, \dots, X_{15}$ denote the first monthly salaries of the group of 15 students. Then, $X_1, \dots, X_{15}$ are i.i.d. $N(2000, 900)$ r.v.'s

- 1 The required prob is given by

$$P(\bar{X} > 2010) = P\left(Z > \frac{2010 - 2000}{\sqrt{900/15}}\right) \approx P(Z > 1.29) = .0985$$

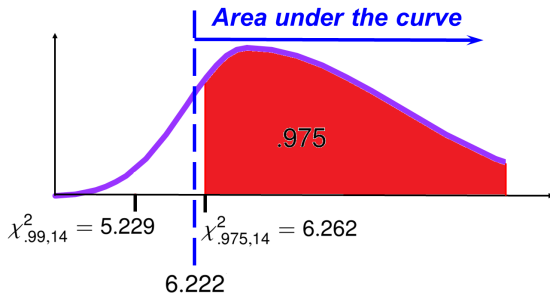
where $\bar{X} = \frac{1}{15} \sum_{i=1}^{15} X_i \sim N(2000, 900/15)$ by (2)

Example 8: Sampling Distributions From a Normal Population II

- 2 The required prob is given by

$$\begin{aligned} P(S > 20) &= P(S^2 > 400) = P\left(\chi_{14}^2 > \frac{(15-1)(400)}{900}\right) \\ &= P(\chi_{14}^2 > 6.222) \in (.975, .99) \end{aligned}$$

where $\frac{(n-1)}{\sigma^2} S^2 = \frac{(15-1)}{900} S^2 \sim \chi_{14}^2$



Example 9: t Statistic From a Normal Population

Consider the last example by assuming that the first monthly salary (in S\$) of a undergraduate in a university is approximately $N(2000, \sigma^2)$ with an unknown $\sigma^2 > 0$. For a selected group of 15 students with sample standard deviation of their first monthly salaries as 35, what is the prob that their average first monthly salary is greater than S\$2010?

Solution: Let $X_1, \dots, X_{15}$ denote the first monthly salaries of the group of 15 students. Then,

① $X_1, \dots, X_{15}$ are i.i.d. $N(2000, \sigma^2)$ r.v.'s, and

② $(\bar{X} - 2000) / \sqrt{S^2/15} \sim t_{15-1}$

The required prob is given by

$$P(\bar{X} > 2010) = P(t_{14} > \frac{2010 - 2000}{\sqrt{35^2/15}}) \approx P(t_{14} > 1.11) > .10$$

as $t_{.10,14} = 1.345$ from the t -table